

2nd Youth International Math Olympiad

Division 2

Test A

Format: Computational problems have answers in the interval $[0, 999]$.
Justification is required for proofs.
Scoring: Each computational question is 3 points, and each proof is 7 points.
Time: 60 minutes

1. Consider $\triangle ABC$ with l as the angle bisector of $\angle A$. Let the perpendiculars from B and C to l intersect l at D and E respectively. Show that the line parallel to AC through D and the line parallel to AB through E intersect at the midpoint of BC .

2. For any positive integer n , let $d(n)$ denote the number of positive divisors of n .

Find the last three digits of the sum of all positive integers $n \leq 2026$ such that

$$d(n^2) = 3 \cdot d(n).$$

3. Define the function

$$f(x) = \frac{x^2 + 1}{2x}$$

for all $x \neq 0$.

The value of $f^{2026}(2)$ can be expressed as $\frac{m}{n}$ for relatively prime positive integers m and n . The value of $m + n$ can be expressed as $a^{(b^c)}$ for positive integers a, b and c with a and b prime. Find the remainder when $a + b + c$ is divided by 1000.

Note: $f^n(x)$ denotes the function f applied n times. For example, $f^2(x) = f(f(x))$.

4. Let n be a positive integer. At a party with n people, each of the n people shakes hands with k other people, where k is some constant positive integer.

In terms of n , find all possible values of k .

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Division 2

Test B

Format: Computational problems have answers in the interval $[0, 999]$.
Justification is required for proofs.
Scoring: Each computational questions is 3 points, and each proof is 7 points.
Time: 60 minutes

1. Two positive integers x and y are considered friendly if there exists a positive integer b such that x and y have the same digit sum when expressed in base b .

Two positive integers x and y are considered distantly friendly if there exists a sequence of positive integers

$$x = a_1, a_2, \dots, a_{k-1}, a_k = y$$

such that a_i and a_{i+1} is friendly for all $1 \leq i \leq k-1$.

Determine the number of pairs of integers $1 \leq a < b \leq 2026$ such that a and b are not distantly friendly.

2. Let $\triangle JFK$ be an acute scalene triangle with circumcenter O and circumcircle ω . Let D be the foot of the altitude from J to FK . The perpendicular bisector of segment JD meets ω at two points X and Y . Let ℓ be the line through O parallel to FK . The circumcircle of triangle XYD meets ℓ at two points P and Q . Prove that the midpoint of PQ is O .

3. Define a sequence $(a_n)_{n \geq 1}$ by

$$a_1 = a_2 = 1,$$

and

$$a_n = \frac{a_{n-1}^2 + 2}{a_{n-2}} \quad \text{for all } n \geq 3.$$

Prove that all terms of the sequence are positive integers.

4. Let $S = \{1, 2, 3, \dots, 2026\}$. A non-empty subset $A \subseteq S$ is called *balanced* if the arithmetic mean of the elements in A is an integer. Let $\mathcal{B}$ be the set of all balanced subsets of S .

Compute the remainder when

$$\sum_{A \in \mathcal{B}} (-1)^{|A|}$$

is divided by 1000, where $|A|$ denotes the number of elements in the subset A .

2nd Youth International Math Olympiad

Division 1

Test A

Format: Computational problems have answers in the interval $[0, 999]$.
Justification is required for proofs.
Scoring: Each computational question is 3 points, and each proof is 7 points.
Time: 60 minutes

1. Let γ be a circle passing through a point P .

Three circles $\omega_1, \omega_2, \omega_3$ all pass through P , and their centers lie on γ . Let A be the second intersection point of ω_1 and ω_2 , let B be the second intersection of ω_2 and ω_3 , and let C be the second intersection of ω_1 and ω_3 .

Prove that A, B and C are collinear.

2. Let the sequence of integers $(x_n)_{n \geq 0}$ be defined by

$$x_0 = 6, \quad x_1 = 18, \quad \text{and} \quad x_{n+1} = 6x_n - x_{n-1} \quad \text{for all } n \geq 1.$$

The highest power of 2 that divides

$$x_{2^{2026}} - 6$$

is 2^x for some positive integer x . Find the remainder when x is divided by 1000.

3. Let a, b, c be positive real numbers such that

$$a^3 + b^3 + c^3 + abc = 4$$

Prove that

$$\frac{a^2}{\sqrt{b+c}} + \frac{b^2}{\sqrt{c+a}} + \frac{c^2}{\sqrt{a+b}} \geq \frac{3}{\sqrt{2}}.$$

4. An unfair coin lands heads with probability $\frac{2}{3}$. It is flipped k times. Let $E(k)$ denote the expected number of integers $1 \leq i \leq k$ at which the number of tails is double the number of heads in the first i flips.

The limit of $E(k)$ as k approaches infinity can be expressed as $\frac{\sqrt{a}-b}{c}$ for positive integers a, b and c with a squarefree. Find $a + b + c$.

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Division 1

Test B

Format: Computational problems have answers in the interval $[0, 999]$.
Justification is required for proofs.
Scoring: Each computational question is 3 points, and each proof is 7 points.
Time: 60 minutes

1. For positive real numbers x, y, z such that

$$x^2 + 4y^2 + 9z^2 = 256,$$

determine the maximum possible value of $(x + y)(y + 3z)$.

2. For each integer $a \in \{1, 2, \dots, 102\}$ let the product $P(a)$ be

$$P(a) = \prod_{k=1}^{102} (k^2 + a).$$

Find the remainder when the sum $\sum_{a=1}^{102} P(a)$ is divided by 103.

3. Let $\triangle ABC$ be an acute scalene triangle with circumcenter O , orthocenter H , and centroid G . Let E and F be the midpoints of CA and AB , respectively.

Let P be the reflection of H across BC , and let Q be the reflection of O across EF .

Part (a): Show that P, G, Q are collinear.

Part (b): Show that Q lies on the nine-point circle of ABC .

4. Let $n \geq 2$ be an integer. A conference has n^2 committees, each containing exactly $2n + 1$ people. Any two distinct committees have at most one member in common.

Prove that every committee can appoint a chair and a vice-chair so that all $2n^2$ appointed officers are different people.